Code No: 123BP **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, February - 2024
DATA STRUCTURES
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART- A

(25 Marks)

1.a)	What is sparse matrix? Show array representation of a sparse matrix.	[2]
b)	What is amortized complexity? Discuss briefly.	[3]
c)	What is an ADT? Write stack ADT.	[2]
d)	What is Deque? Indicate how deque is represented in C.	[3]
e)	Define strictly binary tree. Give example.	[2]
f)	Explain adjacency matrix representation of a graph with example.	[3]
g)	What is Hash table?	[2]
h)	What is sorting? What is searching?	[3]
i)	Define AVL tree.	[2]
j)	What are splay trees?	[3]

PART-B

(50 Marks)

2. Give the node structure to create a linked list of integers and write C functions to perform the following:
a) Create a three node list with data 11, 22, and 33.
b) Insert a node with data value 15 in between the nodes 11 and 22.
c) Display the resulting singly linked list. [3+3+4]

OR

3.a) Explain different types of linked lists with diagram.
b) Write a C program to insert a node into a doubly linked list. [5+5]

4.a) Discuss the various exceptional conditions that should be handled while using stacks.
b) Convert the following postfix expressions to infix expressions:
i) A B C D E / * - F / G + +
ii) a b c d g / * + +. [5+5]

OR

5.a) Define stack and write functions for operations over stack.
b) Write a C program to reverse a string using stack. [5+5]

6. Explain insertion and deletion from a Max Heap with examples. [10]

OR

7.a) Write a C program to implement BFS.

b) Write C functions for the following tree traversals:
i) In order ii) Pre order iii) Post order. [5+5]

8.a) Apply heap sort on the following list of elements:
{14, 12, 9, 8, 7, 10, 18, 20, 30}

b) What is hash function? Explain different hash functions with example. [5+5]

OR

9. Analyze input (371, 325, 163, 199, 344, 679, 989, 534, 448, 355) and the hash function is h(x) = x mod 10. Show the result using quadratic probing and double hashing $h_2(x) = 7 – (x \bmod 7)$. [10]

10.a) Construct a b-tree of order 3 for the following:
{5, 3, 21, 9, 1, 13, 2, 7, 10, 12, 4, 8}

b) Explain about Red-Black trees. [5+5]

OR

11. Explain different types of AVL rotations with example. [10]

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